

2011年 数理方程与特殊函数期末考题答案.

$$1. \begin{cases} \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 & -1 < x < 1, t > 0 & \dots\dots 3 \\ \frac{\partial u}{\partial x}|_{x=-1} = -\frac{2}{k}, \quad \frac{\partial u}{\partial x}|_{x=1} = 0 & \dots\dots 4 \\ u|_{t=0} = \phi(x) & \dots\dots 5 \end{cases}$$

$$2. \begin{cases} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = 0, & r < \rho < R, 0 < \theta < 2\pi & \dots\dots 3 \\ u|_{\rho=r} = 1 & \dots\dots 4 \\ \frac{\partial u}{\partial \rho}|_{\rho=R} = 0 & \dots\dots 5 \end{cases}$$

$$3. \begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} + \frac{R}{\rho} \frac{\partial u}{\partial t} = 0 & 0 < x < 2, t > 0 & \dots\dots 3 \\ \frac{\partial u}{\partial x}|_{x=0} = 0, \quad u|_{x=2} = \sin t & \dots\dots 4 \\ u|_{t=0} = 0, \quad \frac{\partial u}{\partial t}|_{t=0} = 0 & \dots\dots 5 \end{cases}$$

二. 由边界条件是第一类的, 知特征函数系为 $\{\sin nx\}_{n=1}^{\infty}$

$$\text{令 } u(x, t) = \sum_{n=1}^{\infty} u_n(t) \sin nx$$

代入方程和初始条件中得

$$\begin{cases} \sum_{n=1}^{\infty} [u_n'' + a^2 n^2 u_n] \sin nx = \sin 2x \\ u_n(0) = 0 \quad u_n'(0) = 0 \end{cases} \quad \dots\dots 5$$

比较可得

$$u_n'' + a^2 n^2 u_n = 0 \quad n \neq 2$$

$$\text{与 } u_2'' + 4a^2 u_2 = 1 \quad \dots\dots 10$$

因此得 $u_n \equiv 0 \quad n \neq 0$

u_2 的通解 $u_2 = C_1 \cos 2at + C_2 \sin 2at + \frac{1}{4a^2} \dots\dots\dots 13$

由 $u_2(0) = 0$ 和 $u_2'(0) = 0$ 得

$$C_1 = -\frac{1}{4a^2}, \quad C_2 = 0$$

故 $u_2(t) = \frac{1}{4a^2} [1 - \cos 2at]$

因此 $u(x,t) = \frac{1}{4a^2} [1 - \cos 2at] \cdot \sin 2x \dots\dots\dots 15$

三. 令 $w(y) = -\frac{1}{4}y^2$, $v(x,y) = u(x,y) - w(y)$

则 $v(x,y)$ 满足

$$\begin{cases} \frac{\partial^2 v}{\partial x^2} - \frac{\partial^2 v}{\partial x \partial y} - 2 \frac{\partial^2 v}{\partial y^2} = 0 \\ v|_{y=0} = 0, \quad \frac{\partial v}{\partial y}|_{y=0} = 2x \end{cases} \dots\dots\dots 5$$

偏微分方程的特征方程为 $(dy)^2 + dx dy - 2(dx)^2 = 0$

特征线 $y - x = C_1, \quad y + 2x = C_2$

作特征变换 $\xi = y - x, \quad \eta = y + 2x$,

则 $v_{\xi\eta} = 0$

则 $v = f_1(y-x) + f_2(y+2x) \dots\dots\dots 10$

代入定解条件:

$$f_1(-x) + f_2(2x) = 0 \quad ①$$

$$f_1'(-x) + f_2'(2x) = 2x \quad ②$$

②式积分

$$-f_1(-x) + \frac{1}{2}f_2(2x) = x^2 + C \quad ②'$$

① ②' 联立, 得

$$f_1(-x) = -\frac{2}{3}x^2 - \frac{2}{3}C$$

$$f_2(2x) = \frac{2}{3}x^2 + \frac{2}{3}C$$

即

$$f_1(x) = -\frac{2}{3}x^2 - \frac{2}{3}C$$

$$f_2(x) = \frac{1}{6}x^2 + \frac{2}{3}C$$

代入 v 的通式可得

$$\begin{aligned} v &= -\frac{2}{3}(y-x)^2 + \frac{1}{6}(y+2x)^2 \\ &= -\frac{1}{2}y^2 + 2xy \end{aligned}$$

故

$$u = 2xy - \frac{3}{4}y^2 \quad \dots \dots \dots 15$$

四. 对 x 做 Fourier 变换 $u(x, y) \rightarrow U(\omega, y)$ $h(x) \rightarrow H(\omega)$

$$\begin{cases} \frac{d^2 U}{dy^2} + (i\omega)^2 U = 0 & y > 0 \\ U|_{y=0} = H(\omega) \\ \lim_{y \rightarrow \infty} U = 0 \end{cases} \quad \dots \dots \dots 5$$

常微分方程通解 $U = C_1 e^{-|\omega|y} + C_2 e^{|\omega|y} \quad \dots \dots \dots 9$

由 $\lim_{y \rightarrow \infty} U = 0$ 得 $C_2 = 0$

再由 $U|_{y=0} = H(\omega)$ 得 $C_1 = H(\omega)$

故 $U = H(\omega) e^{-|\omega|y} \quad \dots \dots \dots 13$

所以

$$\begin{aligned}
 u(x, t) &= F^{-1} [H(\omega) e^{-|\omega|y}] \\
 &= F^{-1} [H(\omega)] * F^{-1} [e^{-|\omega|y}] \\
 &= h(x) * \frac{1}{\pi} \frac{y}{y^2 + x^2} \\
 &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{y}{y^2 + (x-t)^2} h(t) dt \quad \dots\dots\dots 15
 \end{aligned}$$

五. 取 $M_0(x_0, y_0, z_0)$ 为半空间 $x \geq a$ 内的一点, 则其关于 $x=a$ 平面的对称点为 $M_1(2a-x_0, y_0, z_0)$ ---- 4
故 $x > a$ 上的格林函数为

$$\begin{aligned}
 G(M, M_0) &= \frac{1}{4\pi} \left(\frac{1}{r_{MM_0}} - \frac{1}{r_{MM_1}} \right) \\
 &= \frac{1}{4\pi} \left(\frac{1}{[(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]^{\frac{1}{2}}} - \frac{1}{[(x+x_0-2a)^2 + (y-y_0)^2 + (z-z_0)^2]^{\frac{1}{2}}} \right) \quad \dots\dots\dots 10
 \end{aligned}$$

又在 $x > a$ 上

$$\begin{aligned}
 \frac{\partial G}{\partial n} \Big|_{x=a} &= -\frac{\partial G}{\partial x} \Big|_{x=a} \\
 &= \frac{1}{2\pi} \frac{a-x_0}{[(a-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]^{\frac{3}{2}}} \quad \dots\dots\dots 12
 \end{aligned}$$

故 $u(x_0, y_0, z_0) = -\frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{(a-x_0) \psi(y, z)}{[(a-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]^{\frac{3}{2}}} dy dz \quad \dots\dots\dots 15$

$$\therefore f(x) = 1-x^2 = \sum_{n=1}^{\infty} C_n J_0(\alpha_n x) \quad \text{----- 3}$$

其中

$$C_n = \frac{2}{J_1^2(\alpha_n)} \int_0^1 x(1-x^2) J_0(\alpha_n x) dx$$

$$= \frac{2}{J_1^2(\alpha_n)} \left[\int_0^1 x J_0(\alpha_n x) dx - \int_0^1 x^3 J_0(\alpha_n x) dx \right] \quad \text{---- 6}$$

而

$$\int_0^1 x J_0(\alpha_n x) dx = \left. \frac{x J_1(\alpha_n x)}{\alpha_n} \right|_0^1 = \frac{J_1(\alpha_n)}{\alpha_n} \quad \text{----- 10}$$

$$\begin{aligned} \int_0^1 x^3 J_0(\alpha_n x) dx &= \int_0^1 x^2 d \left[\frac{x J_1(\alpha_n x)}{\alpha_n} \right] \\ &= \left. \frac{x^3 J_1(\alpha_n x)}{\alpha_n} \right|_0^1 - \frac{2}{\alpha_n} \int_0^1 x^2 J_1(\alpha_n x) dx \\ &= \frac{J_1(\alpha_n)}{\alpha_n} - \frac{2}{\alpha_n^2} x^2 J_2(\alpha_n x) \Big|_0^1 \\ &= \frac{J_1(\alpha_n)}{\alpha_n} - \frac{2 J_2(\alpha_n)}{\alpha_n^2} \quad \text{----- 13} \end{aligned}$$

从而

$$C_n = \frac{4 J_2(\alpha_n)}{\alpha_n^2 J_1^2(\alpha_n)}$$

从而

$$1-x^2 = \sum_{n=1}^{\infty} \frac{4 J_2(\alpha_n)}{\alpha_n^2 J_1^2(\alpha_n)} J_0(\alpha_n x) \quad \text{----- 15}$$

七. 定解问题

$$\begin{cases} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0 \\ u|_{\theta=0} = 0, \quad u|_{\theta=\alpha} = 0 \\ \left(\frac{\partial u}{\partial r} + u\right)|_{r=1} = -\theta \end{cases}$$

令 $u(r, \theta) = R(r) \Theta(\theta)$, 则得

$$\begin{cases} \Theta'' + \lambda \Theta = 0 \\ \Theta(0) = \Theta(\alpha) = 0 \end{cases} \quad \begin{cases} r^2 R'' + r R' - \lambda R = 0 \\ |R(0)| < +\infty \end{cases}$$

求解本征值问题

$$\lambda = \frac{n^2 \pi^2}{\alpha^2} \quad \Theta_n(\theta) = \sin \frac{n\pi}{\alpha} \theta$$

$$R_n(r) = r^{\frac{n\pi}{\alpha}}$$

$$\text{则 } u_n(r, \theta) = C_n r^{\frac{n\pi}{\alpha}} \sin \frac{n\pi}{\alpha} \theta$$

$$u(r, \theta) = \sum_{n=1}^{\infty} C_n r^{\frac{n\pi}{\alpha}} \sin \frac{n\pi}{\alpha} \theta$$

$$\text{又 } \frac{\partial u}{\partial r} \Big|_{r=1} = \sum_{n=1}^{\infty} \frac{n\pi}{\alpha} C_n \sin \frac{n\pi}{\alpha} \theta$$

$$u|_{r=1} = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{\alpha} \theta$$

代入圆周边界条件得

$$-\theta = \sum_{n=1}^{\infty} C_n \left(1 + \frac{n\pi}{\alpha}\right) \sin \frac{n\pi}{\alpha} \theta$$

$$\text{故 } C_n = -\frac{1}{1 + \frac{n\pi}{\alpha}} \cdot \frac{2}{\alpha} \int_0^{\alpha} \theta \sin \frac{n\pi}{\alpha} \theta d\theta = \frac{2\alpha^2 (-1)^n}{n\pi(\alpha + n\pi)}$$

$$u(r, \theta) = \frac{2\alpha^2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n(\alpha + n\pi)} r^{\frac{n\pi}{\alpha}} \sin \frac{n\pi}{\alpha} \theta$$